Computer Hardware Full Forms & All Important Shortcut Keys

This is a complete study guide for Computer Hardware, Basic Computer, Windows, MS Office, and Hardware & Networking students. You can use these notes for classroom teaching, exam preparation, and practical revision.

Part 1: Computer Hardware Full Forms

1. Basic Computer Terms

Short Form	Full Form
CPU	Central Processing Unit
GPU	Graphics Processing Unit
PC	Personal Computer
UPS	Uninterruptible Power Supply
SMPS	Switched-Mode Power Supply
PSU	Power Supply Unit
ALU	Arithmetic Logic Unit
CU	Control Unit
MU	Memory Unit
BIOS	Basic Input/Output System
UEFI	Unified Extensible Firmware Interface
POST	Power-On Self-Test
CMOS	Complementary Metal-Oxide-Semiconductor
PCB	Printed Circuit Board
IC	Integrated Circuit
USB	Universal Serial Bus
I/O	Input/Output
OS	Operating System
GUI	Graphical User Interface
CLI	Command-Line Interface

2. Memory and Storage

Short Form	Full Form

RAM	Random Access Memory
ROM	Read-Only Memory
SRAM	Static Random Access Memory
DRAM	Dynamic Random Access Memory
SDRAM	Synchronous Dynamic Random Access Memory
DDR	Double Data Rate
HDD	Hard Disk Drive
SSD	Solid-State Drive
NVMe	Non-Volatile Memory Express
SATA	Serial Advanced Technology Attachment
PATA	Parallel Advanced Technology Attachment
M.2	Form-factor specification for internal expansion and storage devices
SD	Secure Digital
CD	Compact Disc
DVD	Digital Versatile Disc
BD	Blu-ray Disc
TB	Terabyte
GB	Gigabyte
MB	Megabyte
KB	Kilobyte
HDD RPM	Hard Disk Drive Revolutions Per Minute
ECC	Error-Correcting Code

Important: RAM is generally volatile memory, meaning its contents are lost when power is removed. ROM and SSD/HDD storage retain data without power.

3. Motherboard and Processor

Short Form	**Full Form**
FSB	Front-Side Bus
PCI	Peripheral Component Interconnect

PCIe	Peripheral Component Interconnect Express
AGP	Accelerated Graphics Port
SATA	Serial Advanced Technology Attachment
IDE	Integrated Drive Electronics
TPM	Trusted Platform Module
VRM	Voltage Regulator Module
CPU Socket	Processor Socket
DIMM	Dual In-Line Memory Module
SO-DIMM	Small Outline Dual In-Line Memory Module
POST	Power-On Self-Test
BIOS	Basic Input/Output System
UEFI	Unified Extensible Firmware Interface
TDP	Thermal Design Power
GHz	Gigahertz
MHz	Megahertz
NPU	Neural Processing Unit
iGPU	Integrated Graphics Processing Unit
dGPU	Discrete Graphics Processing Unit

4. Display and Graphics

Short Form	**Full Form**
VGA	Video Graphics Array
HDMI	High-Definition Multimedia Interface
DVI	Digital Visual Interface
DP	DisplayPort
LCD	Liquid Crystal Display
LED	Light-Emitting Diode
OLED	Organic Light-Emitting Diode

IPS	In-Plane Switching
TN	Twisted Nematic
HDR	High Dynamic Range
FHD	Full High Definition
QHD	Quad High Definition
UHD	Ultra High Definition
FPS	Frames Per Second
VRAM	Video Random Access Memory
HDCP	High-bandwidth Digital Content Protection

5. Power Supply and Cooling

Short Form	**Full Form**
AC	Alternating Current
DC	Direct Current
UPS	Uninterruptible Power Supply
SMPS	Switched-Mode Power Supply
PSU	Power Supply Unit
PFC	Power Factor Correction
PWM	Pulse-Width Modulation
RPM	Revolutions Per Minute
AIO	All-in-One (liquid cooling system)
V	Volt
W	Watt
A	Ampere
VA	Volt-Ampere
Wh	Watt-hour
kWh	Kilowatt-hour

6. Networking and Internet

Short Form	Full Form
LAN	Local Area Network
WAN	Wide Area Network
MAN	Metropolitan Area Network
PAN	Personal Area Network
WLAN	Wireless Local Area Network
NIC	Network Interface Card
MAC	Media Access Control
IP	Internet Protocol
IPv4	Internet Protocol Version 4
IPv6	Internet Protocol Version 6
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
DNS	Domain Name System
DHCP	Dynamic Host Configuration Protocol
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
FTP	File Transfer Protocol
SFTP	SSH File Transfer Protocol
SSH	Secure Shell
VPN	Virtual Private Network
ISP	Internet Service Provider
WLAN	Wireless Local Area Network
SSID	Service Set Identifier
WPA	Wi-Fi Protected Access
WPA2	Wi-Fi Protected Access 2
WPA3	Wi-Fi Protected Access 3

RJ45	Registered Jack 45 (commonly used name for Ethernet connectors)
PoE	Power over Ethernet
VLAN	Virtual Local Area Network
NAT	Network Address Translation
ARP	Address Resolution Protocol
ICMP	Internet Control Message Protocol
MTU	Maximum Transmission Unit
QoS	Quality of Service
OSI	Open Systems Interconnection
Mbps	Megabits per second
Gbps	Gigabits per second

7. Ports, Connectors and Cables

Short Form	**Full Form**
USB	Universal Serial Bus
USB-C	USB Type-C connector
USB-A	USB Type-A connector
USB-B	USB Type-B connector
HDMI	High-Definition Multimedia Interface
VGA	Video Graphics Array
DVI	Digital Visual Interface
DP	DisplayPort
RJ11	Registered Jack 11
RJ45	Registered Jack 45
PS/2	Personal System/2
SATA	Serial Advanced Technology Attachment
eSATA	External Serial Advanced Technology Attachment
Thunderbolt	Intel/Apple-developed interface technology

TRS	Tip-Ring-Sleeve
TRRS	Tip-Ring-Ring-Sleeve

8. Printer and Peripheral Devices

Short Form	Full Form
DPI	Dots Per Inch
PPM	Pages Per Minute
ADF	Automatic Document Feeder
MFP	Multifunction Printer
OCR	Optical Character Recognition
OMR	Optical Mark Recognition
MICR	Magnetic Ink Character Recognition
POS	Point of Sale
HID	Human Interface Device
TWAIN	Technology Without an Interesting Name

9. Cybersecurity and Hardware Support

Short Form	Full Form
MFA	Multi-Factor Authentication
2FA	Two-Factor Authentication
ACL	Access Control List
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
EDR	Endpoint Detection and Response
AV	Antivirus
DoS	Denial of Service
DDoS	Distributed Denial of Service
CIA	Confidentiality, Integrity, Availability
RDP	Remote Desktop Protocol

RMM	Remote Monitoring and Management
PXE	Preboot Execution Environment
KVM	Keyboard, Video, Mouse
KVM Switch	Keyboard, Video, Mouse switch

Part 2: All Important Computer Shortcut Keys

1. Basic Keyboard Shortcuts

These shortcuts work in many Windows applications, though some vary by program.

Shortcut Key	**Function**
Ctrl + C	Copy selected content
Ctrl + X	Cut selected content
Ctrl + V	Paste content
Ctrl + Z	Undo last action
Ctrl + Y	Redo or repeat an action
Ctrl + A	Select all
Ctrl + S	Save
Ctrl + P	Print
Ctrl + N	Create a new document or window
Ctrl + O	Open a file
Ctrl + F	Find text
Ctrl + H	Find and replace in supported applications
Ctrl + W	Close current document or tab
Ctrl + B	Bold
Ctrl + I	Italic
Ctrl + U	Underline
Ctrl + K	Insert hyperlink in supported applications
Ctrl + E	Center-align text in many editors
Ctrl + L	Left-align text in many editors

Ctrl + R	Right-align text in many editors
Ctrl + J	Justify text in many editors
Ctrl + Home	Move to beginning of document
Ctrl + End	Move to end of document
Home	Move to beginning of current line
End	Move to end of current line
Shift + Arrow	Select text or items
Ctrl + Shift + Esc	Open Task Manager
Alt + F4	Close the active application or window
F1	Help in many applications
F2	Rename selected file or folder in File Explorer
F5	Refresh in many applications
Esc	Cancel or close a menu or dialog in many cases

2. Windows Operating System Shortcuts

Shortcut Key	**Function**
Win	Open Start menu
Win + D	Show or hide desktop
Win + E	Open File Explorer
Win + I	Open Windows Settings
Win + L	Lock computer
Win + R	Open Run dialog
Win + S	Open Windows Search
Win + A	Open Quick Settings in Windows 11
Win + X	Open Quick Link menu
Win + Tab	Open Task View
Win + M	Minimize all windows
Win + Shift + M	Restore minimized windows

Win + Left Arrow	Snap window to left
Win + Right Arrow	Snap window to right
Win + Up Arrow	Maximize window in supported layouts
Win + Down Arrow	Restore or minimize window
Win + P	Choose display projection mode
Win + K	Open Cast/connect panel in supported configurations
Win + U	Open Accessibility settings
Win + V	Open clipboard history (enable it first if needed)
Win + .	Open emoji and symbol panel
Win + H	Open voice typing
Win + Space	Switch input language or keyboard layout
Win + Ctrl + D	Create a virtual desktop
Win + Ctrl + Left/Right Arrow	Switch virtual desktops
Win + Ctrl + F4	Close current virtual desktop
Win + PrtScn	Save a screenshot to the Screenshots folder
Win + Shift + S	Capture a selected screen area
Alt + Tab	Switch between open applications
Ctrl + Shift + Esc	Open Task Manager
Ctrl + Alt + Delete	Open the Windows security screen

3. File Explorer Shortcuts

Shortcut Key	**Function**
Win + E	Open File Explorer
Ctrl + N	Open a new File Explorer window
Ctrl + Shift + N	Create a new folder
Ctrl + L	Select the address bar
Alt + Left Arrow	Go back
Alt + Right Arrow	Go forward

Alt + Up Arrow	Go to the parent folder
F2	Rename selected file or folder
Delete	Move selected item to Recycle Bin, usually
Shift + Delete	Delete without using Recycle Bin in normal cases
Ctrl + A	Select all items
Ctrl + C	Copy selected files
Ctrl + X	Cut selected files
Ctrl + V	Paste copied or cut files
Alt + Enter	Open Properties of the selected item
Ctrl + Mouse wheel	Change icon size in supported views
F5	Refresh current view
Alt + P	Toggle Preview pane in supported versions
Ctrl + Shift + E	Expand the navigation tree to the current folder in supported versions

Practical example: To create a new folder, open the desired location and press Ctrl + Shift + N. Type the folder name and press Enter.

4. Screenshot and Screen Recording Shortcuts

Shortcut Key	Function
PrtScn	Capture the screen or open the screen-capture experience, depending on settings
Win + PrtScn	Save a full-screen screenshot
Win + Shift + S	Open the screen snipping tool
Alt + PrtScn	Capture the active window in supported configurations
Win + G	Open Xbox Game Bar
Win + Alt + R	Start or stop recording with Game Bar, where supported
Win + Alt + PrtScn	Capture a game screenshot with Game Bar, where supported

5. System and Troubleshooting Shortcuts

Shortcut Key	Function
Ctrl + Shift + Esc	Open Task Manager
Win + R, then msconfig	Open System Configuration

Win + R, then devmgmt.msc	Open Device Manager
Win + R, then diskmgmt.msc	Open Disk Management
Win + R, then ncpa.cpl	Open Network Connections
Win + R, then appwiz.cpl	Open Programs and Features
Win + R, then services.msc	Open Services
Win + R, then eventvwr.msc	Open Event Viewer
Win + R, then dxdiag	Open DirectX Diagnostic Tool
Win + R, then msinfo32	Open System Information
Win + R, then control	Open Control Panel
Win + R, then winver	Display Windows version
Win + R, then devmgmt.msc	Check hardware and driver status
Ctrl + Alt + Delete	Open Windows security options

These Run commands are particularly useful for Computer Hardware and Networking technicians. Open Run with Win + R, enter the command, and press Enter.

6. MS Word Shortcut Keys

Shortcut Key	**Function**
Ctrl + N	New document
Ctrl + O	Open document
Ctrl + S	Save document
F12	Open Save As dialog in Word
Ctrl + P	Print document
Ctrl + B	Bold text
Ctrl + I	Italic text
Ctrl + U	Underline text
Ctrl + Shift + >	Increase font size
Ctrl + Shift + <	Decrease font size
Ctrl +]	Increase font size by one point
Ctrl + [	Decrease font size by one point

Ctrl + E	Center align
Ctrl + L	Left align
Ctrl + R	Right align
Ctrl + J	Justify
Ctrl + M	Increase paragraph indent
Ctrl + Shift + M	Decrease paragraph indent
Ctrl + 1	Single line spacing
Ctrl + 2	Double line spacing
Ctrl + 5	1.5 line spacing
Ctrl + Enter	Insert page break
Ctrl + Home	Go to document beginning
Ctrl + End	Go to document end
Ctrl + F	Find text
Ctrl + H	Find and replace
Ctrl + K	Insert hyperlink
Ctrl + Shift + C	Copy formatting
Ctrl + Shift + V	Paste formatting in supported versions
Ctrl + Alt + 1	Apply Heading 1 style
Ctrl + Alt + 2	Apply Heading 2 style
Ctrl + Alt + 3	Apply Heading 3 style
Ctrl + Backspace	Delete the previous word
Ctrl + Delete	Delete the next word
Shift + F3	Change text case
Ctrl + Shift + L	Apply bullets in Word
Ctrl + Z	Undo
Ctrl + Y	Redo

7. MS Excel Shortcut Keys

Shortcut Key	Function
Ctrl + N	New workbook
Ctrl + O	Open workbook
Ctrl + S	Save workbook
Ctrl + P	Print worksheet or workbook
Ctrl + C	Copy cells
Ctrl + X	Cut cells
Ctrl + V	Paste cells
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + A	Select current data region or all, depending on context
Ctrl + 1	Open Format Cells dialog
Ctrl + B	Bold
Ctrl + I	Italic
Ctrl + U	Underline
F2	Edit active cell
F4	Repeat last action or cycle reference types while editing a formula
F5	Open Go To dialog
Ctrl + F	Find
Ctrl + H	Replace
Ctrl + ;	Insert current date
Ctrl + Shift + ;	Insert current time
Alt + =	AutoSum
Ctrl + Arrow	Move to edge of data region
Ctrl + Shift + Arrow	Select to edge of data region
Ctrl + Home	Go to cell A1
Ctrl + End	Go to last used cell

Ctrl + Space	Select entire column
Shift + Space	Select entire row
Ctrl + Plus (+)	Insert cells, rows, or columns
Ctrl + Minus (-)	Delete cells, rows, or columns
Ctrl + Page Up	Switch to previous worksheet
Ctrl + Page Down	Switch to next worksheet
Alt + Enter	Insert a new line within a cell while editing
Ctrl + D	Fill down
Ctrl + R	Fill right
Ctrl + T	Create a table
Ctrl + Shift + L	Toggle filters
Ctrl + `	Show or hide formulas
Ctrl + Shift + $	Apply currency format
Ctrl + Shift + %	Apply percentage format
Ctrl + Shift + #	Apply date format in supported regional settings

Important Excel formulas

Formula	**Purpose**
=SUM(A1:A10)	Add values
=AVERAGE(A1:A10)	Calculate average
=MAX(A1:A10)	Find largest value
=MIN(A1:A10)	Find smallest value
=COUNT(A1:A10)	Count cells containing numbers
=COUNTA(A1:A10)	Count non-empty cells
=IF(A1>=35,"Pass","Fail")	Check a condition
=COUNTIF(A1:A10,">=35")	Count values meeting a condition
=SUMIF(A1:A10,">=35")	Add values meeting a condition
=XLOOKUP(E2,A2:A10,B2:B10)	Find a matching value in modern Excel

8. MS PowerPoint Shortcut Keys

Shortcut Key	Function
Ctrl + N	New presentation
Ctrl + M	Insert new slide
Ctrl + D	Duplicate selected object or slide
Ctrl + S	Save presentation
Ctrl + P	Print
Ctrl + B	Bold
Ctrl + I	Italic
Ctrl + U	Underline
Ctrl + K	Insert hyperlink
Ctrl + G	Group selected objects
Ctrl + Shift + G	Ungroup objects
F5	Start slideshow from beginning
Shift + F5	Start slideshow from current slide
Esc	End slideshow
N or Right Arrow	Next slide during slideshow
P or Left Arrow	Previous slide during slideshow
B	Display black screen during slideshow
W	Display white screen during slideshow
Ctrl + A	Select all objects in the current editing context
Ctrl + Z	Undo
Ctrl + Y	Redo
Alt + F5	Start Presenter View in supported configurations

9. Internet Browser Shortcut Keys

Useful for Chrome, Edge, and many other browsers.

Shortcut Key	Function
Ctrl + T	Open new tab

Ctrl + W	Close current tab
Ctrl + Shift + T	Reopen recently closed tab
Ctrl + Tab	Switch to next tab
Ctrl + Shift + Tab	Switch to previous tab
Ctrl + 1 to Ctrl + 8	Switch to a tab by position
Ctrl + 9	Switch to last tab
Ctrl + L	Select address bar
Ctrl + R	Reload page
Ctrl + Shift + R	Hard reload in browsers that support it
Alt + Left Arrow	Go back
Alt + Right Arrow	Go forward
Ctrl + D	Bookmark current page
Ctrl + J	Open downloads
Ctrl + H	Open browsing history
Ctrl + F	Find text on webpage
Ctrl + P	Print webpage
Ctrl + Plus (+)	Zoom in
Ctrl + Minus (-)	Zoom out
Ctrl + 0	Reset zoom
F11	Toggle full-screen mode
Ctrl + Shift + Delete	Open clear browsing data options
Ctrl + U	View page source in supported browsers
F12	Open Developer Tools in supported browsers

10. Command Prompt (CMD) and Networking Shortcuts

Command / Shortcut	**Function**
Win + R, cmd	Open Command Prompt
ipconfig	Display IP configuration

ipconfig /all	Display detailed network configuration
ipconfig /release	Release an IPv4 DHCP lease
ipconfig /renew	Request a new IPv4 DHCP lease
ipconfig /flushdns	Clear DNS resolver cache
ping 8.8.8.8	Test connectivity to an IP address
tracert google.com	Trace the route to a destination
nslookup google.com	Query DNS records
getmac	Display MAC addresses
hostname	Display computer name
arp -a	Display ARP cache entries
netstat -ano	Display network connections and associated process IDs
systeminfo	Display Windows system information
sfc /scannow	Scan and repair protected Windows system files
chkdsk	Check a disk for file-system errors
tasklist	List running processes
taskkill /PID 1234 /F	Force-terminate the process with that PID; replace 1234 with the actual PID
cls	Clear the Command Prompt screen
exit	Close Command Prompt

Important: Run administrative commands such as sfc /scannow in an elevated Command Prompt when required. Use chkdsk and process-termination commands carefully.

11. Special Keys and Their Functions

Key	**Function**
Enter	Confirm a command or insert a new line
Spacebar	Insert a space
Backspace	Delete the character to the left
Delete	Delete selected item or character to the right
Shift	Type uppercase letters or access alternate key symbols
Ctrl	Modifier used with other keys

Alt	Modifier used with other keys
Tab	Move focus or insert a tab
Caps Lock	Toggle uppercase typing
Num Lock	Toggle numeric keypad mode on supported keyboards
Scroll Lock	Control scrolling behavior in some applications
Print Screen	Capture the screen or invoke a screenshot feature
Pause/Break	Special-purpose key used by some legacy tools
Insert	Toggle insert/overwrite mode in supported applications
Page Up	Move up one screen in supported contexts
Page Down	Move down one screen in supported contexts
Arrow Keys	Move cursor or selection
Esc	Cancel an operation or close a menu